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S 650 FINISHED GRADING AND ENVIRONMENTAL PROTECTION

1. SCOPE OF WORK

The work specified in this section shall cover supply of all labors, materials, transportation, equipment and services necessary for the soil and crushed stone fill work for the final grading and environmental protection work as shown on the approved Bidder's drawings or as directed by the Owner.

2. FINISHED GRADING

2.1 General

Finished grading work shall therefore comprise of filling of the low areas of roads, and other areas with soil and top soil to provide elevation, slopes and drainage as shown on the approved Bidder's drawings or as directed by the Owner.

2.2 Rough Grading

The Bidder shall fill, spread and level the roads and general site areas to elevation grades and slope to elevations as required. The fill material shall consist of graded crushed stone or gravel for the roads. The fill material in the areas where seeding or sodding is indicated on the approved Bidder's drawings shall be soil from a borrow source approved Bidder's by the Owner. If the elevation in the areas where topsoil over soil cover is indicated on the approved drawings is higher than what is required, the areas shall be scraped to the required elevations. The crushed stone fill materials shall be of gradation of "Base Coarse" as specified in specification S-250 Clause 3.2.1. Soil cover shall consist of a 200 mm thick compacted layer of soil free lumps, stones, roots and all other organic materials and shall be from an approved off-site borrow source.

2.3 Final Grading

The Bidder shall perform final grading near the completion of the works or as directed by the Owner. The Bidder shall grade and level the areas to produce a smooth grade free from debris, organic materials, objectionable stones, clods, lumps, pockets or high points, properly drained true to the indicated elevations with smooth transition between different elevations.

3. ENVIRONMENTAL PROTECTION

3.1 Seeding and Sodding

3.1.1 General

The Bidder shall furnish materials and perform seeding operations to produce a uniform stand of healthy grass where indicated on the approved Bidder's drawings as "seeded surfaces" or "seeded topsoil" or as directed by the Owner.

3.1.2 Topsoil

A 100 mm thick layer of topsoil shall be uniformly spread over the areas where seeding or sodding has been specified. Topsoil shall be obtained from an offsite borrow source approved by the Owner and shall be fertile, friable, natural surface soil obtained from a well-drained area. Top soil shall be free of all stone, brush, weeds, stumps, roots or other organic litter. Topsoil shall have 5-20 percent organic matter and an acidity range between pH 5.0 to 8.0 inclusive. Prior to the delivery to the site, the Bidder shall provide a statement regarding the source of the topsoil and shall provide test certificates and mechanical analyses.

3.1.3 Liming and Fertilizing

1) General

The Bidder shall test the soils to determine the proper amount of pH and nutrient adjustment required. Test results and subsequent application rates of lime and fertilizer shall be subject to the approval of the Owner.

2) Liming

Use agriculture ground lime, at the rate of 0.5 kg per square meter, with surface soil before completion of ground preparations. The lime must be applied prior to seeding and may be applied in conjunction with the fertilizer. Immediately after application, the area should be disked, harrowed or rototilled to incorporate the lime or lime-fertilizer into the soil.

3) Fertilizer

Fertilizer shall consist of nitrogen, phosphate and potassium nutrients. Fertilizer shall be applied at such a rate that each hectare will receive the following amounts of available units:

Nitrogen 70 kg

Phosphate (P_2O_5) 115 kg

Potassium (K_2O) 115 kg

Fertilizer can be placed during ground preparation or mixed with and placed with seed and mulch during final seeding.

Condition of fertilizer prior to placing shall be approved by the Owner.

4) Seeding and Mulching

- a. Seed used shall conform to the requirements of the Owner's standards and shall consist of a mixture of the following seeds in the amounts indicated:

Type of seed in mixture	kg per hectare
Fescue (Kentucky 31 or Alta)	60
Red top, solid	24
Clover, alsike (inoculated)	16

- b. Seeding shall be performed in accordance with the Owner's specifications with a maximum depth of planting of 12 mm.
- c. Mulching: Immediately after seeding, seeded areas shall be covered with sprayed asphalt straw mulch.
- d. Except as otherwise specified above, the methods of preparation of seed beds, fertilizing, mulching, seeding, sprinkling, maintaining, repairing and reseeding as required will be at the option of the Bidder. The work shall be considered completed after a uniform and dense stand of healthy perennial grass, free from bare spots and gullies formed by erosion, has been produced.

5) Sodding

- a. The Bidder shall spread topsoil and fertilizer over the area to be sodden as specified in subsections 3.1.2 and 3.1.3 and shall furnish and install sod to produce a healthy grass cover where indicated on the approved Bidder's drawings or as directed by the Owner.
- b. The sod to be used shall be approved by the Owner.
- c. The Bidder shall furnish water, and shall maintain sod including watering and mowing for a period of not less than 90 days after lying, but not earlier than the time the Owner agrees that a healthy growth of grass has been achieved for the area.

3.2 Trees and Shrubs

3.2.1 General

- 1) The Bidder shall furnish plants and shrubs, and perform planting operations to produce healthy plants and shrubs as indicated on the approved Bidder's drawings or as directed by the Owner.
- 2) All work shall be done in the manner normally required in good trade and practice of landscaping and the Bidder must use proper means to provide that all plant materials, approved by the Owner for planting, shall be dug, loaded, moved, unloaded, planted, fertilized, pruned, watered, and maintained in such a manner as to insure their lives for healthy growth in their new locations.
- 3) The Bidder shall perform cleanup work in the work areas daily, and periodically throughout the limits of the project, in order to present a neat appearance of the project at all times.
- 4) All plants shall be inspected and approved at the jobsite by the Owner before they will be eligible for planting. This approval shall not be misconstrued to mean final acceptance. Plant materials damaged enroute to the planting site will be subject to rejection at no cost to the Owner.

3.2.2 Quality and Source of Plant Materials

- 1) All plant materials shall be nursery grown, unless written permission is obtained from the Owner to use selected native stock. This permission will be granted only if the Bidder demonstrates that it is impracticable to obtain nursery grown plant material or if, in the opinion of the Owner, selected native stock will be better suited or superior in quality to that obtained from a nursery.
- 2) If permission is given to use selected native stock, the Owner will inspect the materials before digging and the Bidder shall tag each plant selected (as described herein before) and dig the plants as directed. In the event any selected plant does not show a sufficient root system after being dug, it will be rejected. The handling care and other requirements of collecting native plants will be the same as that described for nursery grown stock.
- 3) All plants shall be root pruned, heavily caned and typical of the species. They must possess a healthy, normal and fibrous root system of sufficient size to insure plant growth. The branch system shall be of normal development and free from disfiguring knots, sun scald injuries, abrasions of the bark, dead or dried wood, broken terminal growth or other objectionable disfigurements. All trees shall have reasonably

straight trunks and shall be well branched and symmetrical in accordance with their natural habits or growth.

3.2.3 Handling and Storage

During handling and storage, the Bidder shall protect the plants from drying out by covering the root system with suitable materials or wetting the foliage if necessary. Balled and burlapped materials shall have the earth ball covered with sawdust, woodchips or other suitable materials and kept in a moist condition. Bare root materials in bundles shall have the bundles opened, plants separated, and then heeled-in in the trenches. Container grown plants shall be kept sufficiently moist to insure their livability.

3.2.4 Delivery and Receipt of Plant Materials

The Bidder shall notify to the Owner at least 24 hours before making any delivery of plant material to the Project. Each shipment shall be accompanied by an invoice showing the sizes and varieties included in that particular shipment.

3.2.5 Inspection

All plant materials are subject to inspection at any time during the life of the contract by an authorized representative of the Owner. Approval of plant material during any of such inspections before or during operations, however, shall not be constructed as final acceptance of the plants involved.

3.2.6 Rejection of Plant Material

Any plant having any of the following unnatural or objectionable features will be rejected: excessive abrasions of the bark; dried out root system; excessive dead wood; dried up wood; excessive sun scald injuries; undeveloped and weak top or roots or both; crooked or one sided development of tops; no straight leaders on trees naturally and normally having them; broken or removed leaders; untrue types or sizes; excessively damaged balls of soil; balls of soil dug from loose soil which could not be expected to properly ball and hold in handling; plants actually dead; and plants not otherwise complying with specification herein.

The rejected material shall be removed from the site and replaced as quickly as possible with new plant material of the same kind, meeting the original requirements, at no additional cost to the Owner.

3.2.7 **Construction Methods**

1) Care and Handling of Plants Before and During Planting

- a. The Bidder shall, in loading, unloading or handling plants, exercise the utmost care to prevent injuries to trunks, limbs, branches and roots. Immediately following delivery and inspection at the site of work, all plants must be cared for in an approved horticultural manner satisfactory to the Owner. Plants must not be allowed to dry out.
- b. If not immediately planted, balled plants must be adequately protected by covering the balls of soil with soil, sawdust, hay or straw and shall be watered frequently. The solidity of the balled plants shall be carefully preserved. They must not be handled by their tops in lifting, moving or setting, but must be handled at all times by their balls of soil.

2) Pruning

If necessary, plant material shall be pruned at the site in accordance with standard modern practice. Cuts over 25 mm in diameter shall be painted with approved tree paint.

3) Location of Plants

Planting stakes will be set by the Bidder at the designated locations of all plants to be approved by the Owner. Each stake shall bear an item number or color code as designated by the Owner to coincide with a particular item number.

4) Planting Pits

Planting pits may be machine or hand dug. Square or Circular pits with vertical or slanted sides shall be dug to the dimensions as required for proper setting of plants

5) Setting Plants

- a. All plants shall be planted in pits. Deep planting shall be avoided and, unless otherwise specified or directed, all plants shall be set at such a level that, after settlement, a normal or natural relationship of the crown of the plant with the ground surface will be established. Each plant to be planted in an individual pit shall be planted in the center of the pit, unless otherwise directed. When balled and burlapped plants are set, topsoil shall be carefully tamped under and around the base of each ball to fill all voids.

- b. All plants shall be planted in topsoil which shall be thoroughly settled by watering and taping as to eliminate settling and leaning of the plant material.
- c. When plants are to be grouped together in a plant bed or in a line approximately 1.6 m, or less from each other, the entire area shall be loosened and lumps broken to a minimum depth of 150 mm prior to excavating plant pits.

6) Top soil

Top soil shall be as specified in Subsection 3.1.2 of this Specification.

7) Fertilizer

The Bidder shall furnish and place commercial fertilizer. The fertilizer shall be spread uniformly on the excavated material in a circle at the outer edge of the planting pit. The fertilizer shall be mixed with the excavated soil before backfilling operations are begun.

8) Backfilling

- a. All plants shall be plumb and set at the proper depth. Suitable material removed from the excavation shall be mixed with fertilizer and the pit shall be backfilled as directed. Care shall be exercised in placing backfill under the sides and over the ball so as not disturb same. Backfill to 3/4 of the depth of the ball on all sides, then water uniformly on all sides of the ball in order to eliminate voids and air spaces and allow settlement of the plant. Plants which settle or lean, before or after watering, shall be straightened, raised, replaced or the condition otherwise remedied. Excavated material considered by the Owner to be of unsuitable planting quality shall not be used in the backfill, but shall be replaced with topsoil by the Bidder as directed by the Owner, at no addition cost to the Owner.
- b. All excavated material not used as a backfill, except soils which contain stones, broken concrete or other foreign matter, shall be disposed of in areas as directed by the Owner. Soils containing stone, broken concrete or other foreign matter shall be disposed to off site by the Bidder to the satisfaction of the Owner.

9) Water

The Bidder shall furnish and apply water in sufficient quantities for proper irrigation of the plants. The plants shall be watered during

planting operations as required hereinbefore, immediately after planting and at intervals as directed by the Owner until final acceptance.

- 10) Hydrant system for watering shall be included in this environmental protection scheme and sufficient amount of sprinkle taps and garden hoses shall be furnished in the area required or as directed by Owner.

- 11) Bed Preparation

Unless otherwise directed by the Owner, where the Bid Drawings require locating planting beds, the planting bed areas shall be scraped removing all grass, weeds, sticks, stones, bricks, mortar, concrete and other debris. Beds shall then be tilled with a rototiller to the required depth. Peat moss or an organic matter satisfactory to the Owner, and commercial fertilizer shall then be added to the soil at the following minimum rates: peat moss or organic matter, 1.5 kg pounds per square meter, and fertilizer 0.25 kg per square meter. After the peat moss and fertilizer have been added, beds shall be tilled again to thoroughly mix all components. Beds shall then be raked smooth and all foreign matter such as dirt lumps, concrete, sticks, grass, mortar, brick, etc., removed from the soil. Finish grade of beds shall be the same as finish grades for an adjoining concrete walks. Herbicides used in bed areas shall be approved types by the Owner.

- 12) Mulching

Unless otherwise specified, any approved material suitable for mulch may be used. Mulching shall be accomplished as soon as possible after the plants are set. The mulch shall be placed uniformly on all sides of the plants and watered as soon as possible. When plants are planted in a row approximately 1.5 m apart or less, the entire row shall be mulched solid. Bed areas shall also be mulched solid.

- 13) Weeding

Weeds shall be kept out from within the planting basin of each plant including basin walls. The Bidder shall mow around all plants materials not planted in beds for a distance of 1.5 m. If the Bidder decides to use selective herbicides, he will be allowed to do so only after the grass in this 3 m circle has been cut to height of not more than 50 mm. these areas must be weeded by hand. Weeding shall be performed as required to maintain a neat appearance as directed.

3.2.8 **Period of Establishment and Replacements**

- 1) Upon the completion of planting of all plants, and providing all plants are in place, living and conform to these specifications, the Owner will

give provisional acceptance. The Bidder shall be required to care for all planted areas for a period of establishment, which shall be one full growing season after the provisional acceptance is made. A growing season shall be for a period of six months (during this period of establishment the Bidder shall be required to employ all possible means to preserve the plants in healthy growing condition). Such plant establishment work shall include cultivation, watering, pruning, controlling insect pests and disease, and any other work necessary to insure healthy plant growth. Dead or unsatisfactory plants shall be promptly removed from the site at no additional cost to the Owner.

- 2) All dead and unsatisfactory plants shall be replaced in kind, quantity and size with live, healthy plants installed as original specified. Alternate or substitute varieties of plants shall be used only if approved by the Owner. These replacement plantings shall be made at the Bidder's expense. Replacements made at this time will not be required a period of establishment.
- 3) Upon completion of the said replacement by the Bidder and prior to final acceptance of the Contract, the Bidder shall weed around all plants and remove all discarded materials, rubbish and equipment from within the areas affected by operations under this Contract. The said areas shall present a neat and acceptable appearance at the time of final inspection.
- 4) A final inspection of all plant material will be held approximately two weeks after the replacement planting has been completed. A final acceptance will be made only if all plants are in place, in conformance with the approved Bidder's drawing plans and the Specifications.

3.3 Gravel Pavement

Gravel pavement shall be 200 mm in finished thickness for switchyard and 100 mm for another area, and consist of a base layer and a finish layer. The top level of gravel surfacing shall be EL+2.8m. Material for the base layer shall be crushed stone of a clean, hard and durable quality. Before the placing of the Gavel Pavement, the surface of the sub-grade shall be cleaned of all objectionable substances and properly shaped and drained. Spread material shall be compacted by means of rammer tapping machine or approved equal equipment. The crushed stone shall be of following gradation (Size S-40 JIS A5001), or equivalent of it.

Name of crushed stone: single-sized crushed stone S-40

Sieve Size (mm)	Percentage Passing by Weight (%)
53	100
37.5	85 ~ 100
31.5	0 ~ 15

4. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 4.1. Quality control shall conform to the requirements of Part 4, Subsection 2.7.
- 4.2. The quality control procedures and documentation.

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